Problem 1

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Problem 1 *Consider the following limit of Riemann sums of a function g on $[a, b]$:*

$$\lim_{n\to\infty} \sum_{k=1}^{n} (x_k^* + \cos(x_k^*))\Delta x \ , \ [0, \pi].$$

Express the limit as a definite integral. Use geometry to evaluate the resulting definite integral.

Explanation. We have

$$\int_0^{\pi} (x + \cos(x))\, dx = \lim_{n\to\infty} \sum_{k=1}^{n} (x_k^* + \cos(x_k^*))\Delta x.$$

To evaluate this definite integral, we'll start by writing $\int_0^{\pi} (x + \cos(x))\, dx = \int_0^{\pi} x\, dx + \int_0^{\pi} \cos(x)\, dx$.

The first term is the definite integral $\int_0^{\pi} x\, dx$ which is the net area under the graph of $y = x$ on the interval $[0, \pi]$. This is a triangle of base π and height π, so it has area $\frac{1}{2}\pi^2$.

The second term $\int_0^{\pi} \cos(x)\, dx$ is the net area between the graph of $y = \cos(x)$ and the x-axis on the interval $[0, \pi]$. In the interval $[0, \frac{\pi}{2}]$ the function is positive, and in the interval $[\frac{\pi}{2}, \pi]$ the function is negative. By the symmetry of the cosine function, these two regions have the same geometric area, so the net area is zero.
$\int_0^{\pi} (x + \cos(x))\, dx = \int_0^{\pi} x\, dx + \int_0^{\pi} \cos(x)\, dx = \frac{\pi^2}{2}$.